

Gamma-Shaped Dealer-Book Pressure and Endogenous Volatility Cycles:

A Reduced-Form Fixed-Equilibrium Hopf Model and a Pre-Extraction Predictive Protocol

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Abstract

We study a reduced-form three-state mechanism in which gamma-shaped option-book pressure interacts with a filtered price–variance loop and can generate an endogenous cycle near a fixed reference state. Under the physical measure, the states are detrended log price, instantaneous variance with square-root diffusion, and an exponentially filtered price signal. Centering the book-pressure functional leaves the equilibrium unchanged as coupling varies, while scaling memory feedback by variance preserves inward drift at zero variance. For any three-times continuously differentiable book functional, we derive the cubic Routh–Hurwitz conditions for a Hopf bifurcation, the exact eigenvalue crossing speed, and the first Lyapunov coefficient governing the stability of emerging cycles. A Gaussian density of latent signed positions yields a quadratic threshold equation. In a deliberately noncalibrated example, direct integration of the nonlinear drift produces an attracting cycle whose local amplitude follows the predicted square-root scaling. Signed-mixture checks show strong dependence on book shape and orientation: some mixtures are subcritical, and reversing the canonical orientation removes positive local Hopf roots. The option kernel shapes a postulated pressure term; it does not derive hedge flow or market impact, and the price-to-variance channel is a separate modeling assumption. The numerical result therefore establishes internal feasibility for the specified loop rather than an S&P 500 calibration. A separate preregistered protocol uses public open interest to study participant-sign-free predictive associations before proprietary-data extraction. Dealer inventory and causal effects lie outside its estimand, and the registered market data have not been accessed.

1 Introduction

Option gamma measures how an option’s delta changes with the underlying price. Delta hedging therefore turns an option position into state-dependent demand for its underlying. When the intermediary sector is net long gamma, rebalancing is locally contrarian; when it is net short gamma, it is locally trend-following. This observation motivates a possible feedback channel from option inventories to returns and volatility (Frey and Stremme, 1997; Sircar and Papanicolaou, 1998; Platen and Schweizer, 1998). Demand-based option pricing establishes that intermediary balance sheets and end-user demand can affect *option prices* (Gârleanu et al., 2009); it does not, by itself, identify a drift term for the underlying. Our spot-feedback equation is therefore a reduced-form hypothesis. Testing its signed structural channel would require participant-tagged positions and an identification strategy; the protocol here asks a weaker predictive question using public book summaries.

Here “gamma-shaped pressure” means a postulated price-pressure state whose dependence on price and variance follows an integrated Black–Scholes gamma kernel. Mapping that state to observed hedge flow remains outside the construction.

This paper studies the resulting local mechanism subject to three design requirements. The reference equilibrium remains fixed as coupling changes; the variance drift remains inward-pointing at zero; and the empirical protocol keeps public-book measurements distinct from latent signed dealer inventory. These safeguards prevent a moving equilibrium, an inadmissible variance state, or a measurement convention from creating a spurious result. The question is conditional: can gamma-shaped book pressure destabilize the specified log-price–memory–variance loop? Structural intermediary optimization and a deterministic explanation of observed volatility clustering remain separate research questions.

The paper makes four contributions.

1. We state a physical-measure model for detrended log-price deviations. Centering the dealer-book functional keeps the reference equilibrium fixed for every coupling value. Making memory feedback proportional to variance leaves the variance drift positive at zero.
2. We derive the complete cubic Routh–Hurwitz test, including all positivity side conditions and an exact expression for transversality. We also evaluate the first Lyapunov coefficient, which determines whether the nearby small cycles are attracting or repelling.
3. We specialize the dealer book to a centered Gaussian signed-position density. The Hopf equation becomes a quadratic in κ , while the first Lyapunov coefficient is evaluated from analytic second and third derivatives. The numerical example integrates this actual nonlinearity rather than a separately chosen cubic normal form. We then assess normal-form scaling, numerical tolerances, initial conditions, one-at-a-time parameter changes, the book-shape plane, and signed Gaussian mixtures.
4. We delimit the estimand available from a future proprietary-data exercise. The revised fixed-horizon protocol uses participant-sign-free public-book summaries to study incremental predictive content while reserving dealer-inventory and causal interpretations for participant-tagged data and a separate identification strategy.

The rest of the paper is organized as follows. Section 2 positions the result relative to hedging-feedback, option-demand, endogenous-cycle, and empirical gamma work. Section 3 states the economic object and measurement distinction. Section 4 gives the local analysis, and sections 5 and 6 report the nonlinear and robustness checks. Section 7 fixes the future empirical protocol. The appendices derive the Gaussian book, list all derivatives used in ℓ_1 , and record the numerical derivation and empirical reporting plan.

2 Position in the literature

Hedging feedback. The closest theoretical ancestors derive price or volatility feedback from the demand of dynamic hedgers. Frey and Stremme (1997), Sircar and Papanicolaou (1998), and Platen and Schweizer (1998) show that replication demand need not be price-taking when the hedging sector is large. Those models supply the economic reason that gamma-sensitive rebalancing can alter the underlying process. More recent work studies explicit impact, optimal hedging, and market quality. Anderegg et al. (2022) combine a permanent-impact model with reconstructed FX option-market-maker exposures; Buis et al. (2024) add a dynamic hedger to a simulated limit-order

market. Building on this economic motivation, we study a different object: a fixed-equilibrium local bifurcation with a positive-variance state and a fully differentiated nonlinear book.

Option demand and dealer balance sheets. Demand-based option pricing links end-user demand and intermediary constraints to option prices (Gârleanu et al., 2009). Section 3 introduces a separate reduced-form physical-measure price-pressure channel. Its scale summarizes market depth, hedge frequency, and the fraction of positions actively rebalanced; a structural derivation would additionally require intermediary optimization and market clearing.

Endogenous financial cycles. Hopf and Neimark–Sacker bifurcations are established mechanisms in heterogeneous-agent asset-pricing models (Brock and Hommes, 1998; He et al., 2009; Brock et al., 2009). Moving-average rules provide a memory channel in continuous-time financial models (He and Zheng, 2010). These precedents establish endogenous oscillation as background. The distinct object here is the combination of a filtered price state, a variance state, and a centered signed option-book functional for which the coupling can be varied without changing the reference equilibrium.

Empirical gamma positioning. Empirical studies generally require an inventory reconstruction. Barbon and Buraschi (2020) and Dim et al. (2024) study gamma-related price or volatility effects using positioning assumptions or market-maker inventory measures, while Baltussen et al. (2021) connect intraday momentum to gamma-sensitive hedging demand across futures markets. Vasquez et al. (2025) use participant-tagged information to estimate market-maker positions. The contrast is central to our design: a convention applied to public open interest is not observationally equivalent to participant-tagged inventory. Evidence from FX trade-repository data (Anderegg et al., 2022) is informative about the hedging channel but does not solve the measurement problem for public open interest in S&P 500 Index (SPX) options.

Scope of the contribution. The contribution lies in the conjunction of five auditable elements: a centered and boundary-compatible local system; complete cubic side conditions and transversality; analytic derivatives through third order; direct nonlinear and mixture checks; and an empirical protocol that keeps its observable estimand separate from the latent theoretical book.

3 Economic object, measure, and state variables

All dynamics are under the physical measure $\mathbb{P}$. Let $S_t > 0$ be the underlying index level and let F_t be a positive, predictable reference level that absorbs the baseline expected return, and define the local detrended deviation

$$X_t = \log(S_t/F_t).$$

The analysis concerns a neighborhood of $X = 0$ over a horizon on which the option book is frozen; the model describes local deviations rather than the global price level. The state v_t is annualized instantaneous variance, χ_t is an exponentially weighted filter of X_t , and $g(X_t, v_t, \chi_t)$ is a dimensionless dealer-book pressure functional specified below. Let W^S and W^v be standard Brownian motions under $\mathbb{P}$, with instantaneous correlation ρ . The proposed dynamics are

$$dX_t = \left[-\delta X_t - \frac{1}{2}(v_t - \theta_v) + \kappa g(X_t, v_t, \chi_t) \right] dt + \sqrt{v_t} dW_t^S, \quad (1)$$

$$dv_t = [\kappa_v(\theta_v - v_t) + \gamma v_t \chi_t] dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$d\chi_t = \alpha(\beta X_t - \chi_t) dt, \quad d\langle W^S, W^v \rangle_t = \rho dt. \quad (3)$$

We take $\delta, \kappa_v, \theta_v, \alpha, \beta > 0$, $\gamma, \kappa, \xi \geq 0$, and $\rho \in [-1, 1]$. The parameter δ controls local pull toward the reference price; κ_v and θ_v are the variance mean-reversion rate and level; α is the memory-decay rate; and β maps price deviations into the filtered signal. The coefficient γ maps that signal into variance drift, while κ scales book pressure in the price drift. The symbol γ is a variance-memory gain and is unrelated to the option Greek gamma. Finally, ξ is volatility of variance and ρ is the instantaneous correlation between the price and variance shocks.

With time measured in years, X, χ, β , and g are dimensionless; v and θ_v have units yr^{-1} ; $\delta, \kappa_v, \alpha, \gamma, \kappa$, and ξ have units yr^{-1} ; and ρ is dimensionless. The term $-\frac{1}{2}(v - \theta_v)$ is the Itô correction relative to the reference variance. We take the stochastic equation in the X coordinate as primitive and define its deterministic skeleton by suppressing the Brownian terms in that coordinate. A nonlinear state transformation performed first would generally change this skeleton through Itô's formula. In the nested limit $\delta = \gamma = \kappa = 0$, after restoring the predictable reference drift, (X, v) has Heston-type log-return and Cox–Ingersoll–Ross (CIR) variance dynamics (Heston, 1993). Setting $\kappa = 0$ alone retains the $X \rightarrow \chi \rightarrow v$ channel. We therefore call $\kappa = 0$ the zero-book-pressure restriction rather than the Heston restriction.

The variance feedback is proportional to v . Writing $b_v(v, \chi)$ and $\sigma_v(v)$ for the variance drift and diffusion coefficient, respectively, the boundary values are

$$b_v(0, \chi) = \kappa_v \theta_v > 0, \quad \sigma_v(0) = 0,$$

independently of χ . Thus the added drift is compatible with the non-negative variance state space. The usual Feller inequality $2\kappa_v \theta_v \geq \xi^2$ is a separate sufficient boundary condition for the unperturbed CIR backbone (Feller, 1951); we do not use it in the deterministic bifurcation theorem. Numerical simulation of the stochastic system should still use a positivity-preserving scheme rather than silently evaluating $\sqrt{v}$ at a negative Euler iterate (Lord et al., 2010).

Assumption 1 (Local frozen-book experiment). *Over the theoretical horizon, F_t is predictable, the position density and effective maturity are frozen, and g is C^3 on an open neighborhood of $(0, \theta_v, 0)$. Parameters other than κ are held fixed when the bifurcation parameter varies. The deterministic orbit considered below remains inside that neighborhood and satisfies $v > 0$.*

This assumption defines a local comparative static that isolates the feedback geometry. Allowing the SPX book to evolve would introduce additional state variables and could change both the threshold and the bifurcation type. The autonomous frozen-book system is therefore a constructive example; an actual fixed-maturity book would age during the reported multiweek period, while a rolling-book interpretation would require an explicit replenishment model.

3.1 From hedge demand to a reduced-form state feedback

For intuition, fix a reference forward $F_\star$, define $k = \log(K/F_\star)$ and $K(k) = F_\star e^k$, and let $q(k)$ be an intermediary's signed option-position density, with $q(k) > 0$ denoting a long option position. Suppose the book has common effective maturity T . The intermediary offsets its option delta with the underlying position

$$h(S, v) = - \int q(k) \Delta^{BS}(S, K(k), v, T) dk.$$

Here Δ^{BS} is Black–Scholes delta evaluated using v as the variance input. Holding this variance input and the option book fixed over an infinitesimal price change gives

$$\frac{\partial h}{\partial S} = - \int q(k) \Gamma^{BS}(S, K(k), v, T) dk.$$

Thus a long-gamma intermediary sells underlying after a price increase, whereas a short-gamma intermediary buys. More generally, changes in h also contain variance, time, and inventory terms; schematically,

$$dh = - \int q \Gamma^{BS} dS - \int q \partial_v \Delta^{BS} dv - \int \Delta^{BS} dq + \dots$$

The centered gamma level used below is neither dh nor dh/dt . It restricts the shape of a postulated price-pressure state. Translating hedge demand into the calendar-time drift κg additionally requires a rebalancing rate, a price-impact mapping, other order flow, and contemporaneous implied-volatility dynamics. The reduced form summarizes those ingredients through κ and the state dependence of g ; it does not solve hedge trades, price impact, and market clearing jointly.

The diffusion coefficients do not enter the deterministic Hopf threshold because the threshold is a property of the drift skeleton. They remain economically important: the full stochastic system has state-dependent diffusion, its paths do not converge to a deterministic cycle, and the deterministic threshold need not equal a stochastic bifurcation threshold.

3.2 Measurement boundary: signed dealer positions versus open interest

Let $q(k)$ denote a latent density of *signed dealer positions* at log moneyness $k = \log(K/F_*)$. A positive value means the dealer sector is long the option under the chosen normalization. Every outstanding option has a long and a short side. Public open interest (OI) records the number of outstanding contracts; it does not label the dealer side or even the identity of either holder (Options Industry Council, 2025). Consequently, an empirical open-interest grid cannot be substituted for q without an additional, untestable sign assignment. Recent work that estimates option-market-maker gamma uses proprietary participant-tagged trade data precisely to recover positions (Vasquez et al., 2025).

Proposition 1 (Non-identification of signed dealer inventory from aggregate OI). *Fix contract-level public open interest n_i . Without participant labels or restrictions on which participant type holds each side, the signed dealer position vector is not identified: there exist allocations with the same $(n_i)_i$ and opposite dealer signs.*

Proof. Each of the n_i outstanding contracts contributes one long and one short position. An allocation assigning dealers to the long side and non-dealers to the short side has the same public count as the allocation obtained by swapping those participant labels. Dealer net position changes sign while open interest is unchanged. Applying the construction contract by contract proves that the observation map is not injective. $\square$

This nonidentification follows from the content of the observed field: adding dates of the same public OI measure cannot recover dealer sign without additional information or restrictions.

For theory, let μ_q and $\sigma_q > 0$ denote the mean and dispersion of a positive unit-mass Gaussian shape over fixed log moneyness,

$$\varphi_q(k) = \frac{1}{\sqrt{2\pi\sigma_q^2}} \exp \left\{ -\frac{(k - \mu_q)^2}{2\sigma_q^2} \right\},$$

and let T , r , and q_d denote a common effective maturity, continuously compounded risk-free rate, and continuous dividend yield. Integrating Black–Scholes spot gamma (Black and Scholes, 1973),

and absorbing gross position mass and a common impact scale into κ , gives

$$\mathcal{G}(X, v) = \frac{\exp(-q_d T - X)}{\sqrt{2\pi\tau^2(v)}} \exp\left[-\frac{(X - m(v))^2}{2\tau^2(v)}\right], \quad (4)$$

$$\tau^2(v) = \sigma_q^2 + vT, \quad m(v) = \mu_q - (r - q_d + v/2)T.$$

The functional used in (1) is centered and normalized:

$$g(X, v, \chi) = s \frac{\mathcal{G}(X, v) - \mathcal{G}(0, \theta_v)}{\mathcal{G}(0, \theta_v)}, \quad s \in \{-1, +1\}. \quad (5)$$

The sign s is a latent dealer-position orientation, not an open-interest statistic. More general C^3 functionals, including mixtures and a dependence on χ , fit the theorem below provided $g(0, \theta_v, 0) = 0$. The Gaussian specialization is intentionally static in χ so that its assumptions are auditable. Section A derives (4) by completing the square rather than treating it as an assumed response surface.

Equation (4) also makes a strong low-dimensional closure: the same state v is inserted as one flat Black-Scholes variance for every strike at the effective maturity. The physical-measure variance state does not generally equal option-implied variance, and a volatility smile would add further states or a surface. This closure provides a closed-form representation. A pricing application would require distinct physical and implied-variance states, an implied-volatility surface, and an explicit variance-risk-premium specification.

Remark 1 (What centering does). *Centering subtracts the level of the book functional at the reference state while preserving its local price and variance sensitivities. At any fixed coupling, the removed constant can be absorbed into the predictable reference drift. Subtracting it explicitly prevents the reference equilibrium from moving as κ varies, so the bifurcation parameter traces a family of vector fields with one common equilibrium.*

Proposition 2 (Positive normalization invariance). *Let $\tilde{g} = cg$ for a constant $c > 0$ and $\tilde{\kappa} = \kappa/c$. Then the vector field, Hopf frequency, crossing direction, and first-Lyapunov sign at corresponding parameter values are unchanged. In particular, the numerical magnitude of κ^* cannot be interpreted without the normalization of g .*

Proof. Every occurrence of the book in the vector field is through the product κg . Consequently all state derivatives of the drift at $(\tilde{\kappa}, \tilde{g})$ equal those at (κ, g) . The Jacobian and Taylor tensors B, C are identical, so all stated local quantities agree. $\square$

4 Fixed-equilibrium local Hopf bifurcation

Remove the Brownian terms and write the deterministic drift as $f(X, v, \chi; \kappa)$. Because $g(0, \theta_v, 0) = 0$,

$$z_\star = (0, \theta_v, 0) \quad (6)$$

is an equilibrium for every $\kappa \geq 0$. Put

$$G_X = \partial_X g(z_\star), \quad G_v = \partial_v g(z_\star), \quad G_\chi = \partial_\chi g(z_\star).$$

The Jacobian is

$$J(\kappa) = \begin{pmatrix} a & b & d \\ 0 & -\kappa_v & c \\ m & 0 & -\alpha \end{pmatrix}, \quad \begin{aligned} a &= -\delta + \kappa G_X, & b &= -\frac{1}{2} + \kappa G_v, & d &= \kappa G_\chi, \\ c &= \gamma \theta_v, & m &= \alpha \beta. \end{aligned} \quad (7)$$

For $\det(\lambda I - J) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$, direct expansion gives

$$c_2 = \kappa_v + \alpha - a, \quad (8)$$

$$c_1 = \kappa_v\alpha - a(\kappa_v + \alpha) - dm, \quad (9)$$

$$c_0 = -a\kappa_v\alpha - bcm - dm\kappa_v. \quad (10)$$

The equilibrium is linearly asymptotically stable exactly when $c_2 > 0$, $c_1 > 0$, $c_0 > 0$, and

$$H(\kappa) := c_1(\kappa)c_2(\kappa) - c_0(\kappa) > 0. \quad (11)$$

A prime denotes differentiation with respect to κ .

Theorem 1 (Local Hopf criterion). *Suppose f is C^3 jointly in state and κ near (z_*, κ^*) , with $\kappa^* > 0$. If*

$$c_2(\kappa^*) > 0, \quad c_1(\kappa^*) > 0, \quad c_0(\kappa^*) > 0, \quad H(\kappa^*) = 0, \quad H'(\kappa^*) \neq 0, \quad (12)$$

then $J(\kappa^)$ has a simple pair $\lambda_{\pm} = \pm i\omega^*$ and a negative real eigenvalue, where*

$$\omega^* = \sqrt{c_1(\kappa^*)}, \quad \lambda_3 = -c_2(\kappa^*).$$

The crossing speed is

$$\left. \frac{d \operatorname{Re} \lambda_+}{d\kappa} \right|_{\kappa^*} = -\frac{H'(\kappa^*)}{2\{c_1(\kappa^*) + c_2(\kappa^*)^2\}}. \quad (13)$$

If, in addition, the first Lyapunov coefficient $\ell_1 \neq 0$, a non-degenerate local Hopf bifurcation occurs. In the convention of Kuznetsov (2004), $\ell_1 < 0$ gives stable small-amplitude periodic orbits on the side where the equilibrium pair has positive real part; $\ell_1 > 0$ gives unstable orbits on the opposite side.

Proof. At $H = 0$ with the stated positivity conditions, the cubic factors as $(\lambda + c_2)(\lambda^2 + c_1)$, proving the eigenvalue statement. Implicitly differentiate $P(\lambda, \kappa) = 0$ at $\lambda = i\sqrt{c_1}$ and take real parts. Substitution of $c_0 = c_1c_2$ yields (13). The non-zero derivative and $\ell_1 \neq 0$ are the transversality and non-degeneracy conditions of the classical Hopf theorem (Liu, 1994; Kuznetsov, 2004). $\square$

The theorem is local to a neighborhood of z_* . A negative discriminant for the quadratic below excludes this particular complex-pair crossing, while stability over all couplings additionally requires checking real eigenvalue crossings and other invariant sets. Similarly, a subcritical Hopf supplies a local unstable periodic orbit; hysteresis would require a separate global analysis establishing coexistence and a fold of cycles.

4.1 Static book: a quadratic threshold equation

For (5), $G_\chi = 0$. Define

$$A = \alpha + \kappa_v, \quad M = \alpha\kappa_v, \quad L = (\gamma\theta_v)(\alpha\beta).$$

The product L is the linear gain around the directed loop $X \rightarrow \chi \rightarrow v \rightarrow X$; the last link is supplied by b in (7). The $X \rightarrow \chi \rightarrow v$ channel is an independent modeling assumption, not a consequence of option gamma. In the canonical example the coupling changes the $v \rightarrow X$ link mainly through G_v .

Proposition 3 (A closed three-state loop is necessary in the static-book model). *If $L = 0$, the eigenvalues are $a, -\kappa_v, -\alpha$ for every κ and no non-zero-frequency Hopf bifurcation is possible.*

Proof. When $\gamma\theta_v = 0$ or $\alpha\beta = 0$, one of the two off-diagonal links $v \leftarrow \chi$ and $\chi \leftarrow X$ vanishes. The Jacobian is triangular after a permutation, and its characteristic polynomial factors as $(\lambda - a)(\lambda + \kappa_v)(\lambda + \alpha)$. $\square$

Substitution into (11) gives

$$H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0, \quad (14)$$

where

$$\begin{aligned} A_2 &= G_X^2 A, \\ A_1 &= G_v L - G_X A(A + 2\delta), \\ A_0 &= A(M + \delta A + \delta^2) - \frac{1}{2}L. \end{aligned} \quad (15)$$

At $\kappa = 0$, $c_2 = A + \delta > 0$, $c_1 = M + \delta A > 0$, and $c_0 = \delta M + L/2 > 0$. Hence the zero-book-pressure equilibrium is stable exactly when

$$L < 2A(M + \delta A + \delta^2). \quad (16)$$

This inequality is checked separately in the numerical configuration; positivity of the primitive parameters alone does not imply it.

Each positive real root must be evaluated against $c_2 > 0, c_1 > 0, c_0 > 0$ and (13); positivity of a quadratic root alone is insufficient. The derivatives G_X and G_v follow analytically from (4). Because $\mathcal{G}(0, \theta_v) > 0$, derivatives of the centered functional are simply s times the raw derivatives divided by that value.

To classify a valid Hopf point (z_*, κ_H) , write $J = J(\kappa_H)$ and expand

$$f(z_* + x; \kappa_H) = Jx + \frac{1}{2}B(x, x) + \frac{1}{6}C(x, x, x) + O(\|x\|^4),$$

where $B = D_z^2 f(z_*; \kappa_H)$ and $C = D_z^3 f(z_*; \kappa_H)$ are the symmetric bilinear and trilinear derivative tensors. To distinguish the eigenvector from the position density $q(k)$, let u and p be right and left eigenvectors satisfying $Ju = i\omega u$, $J^\top p = -i\omega p$, and $\langle p, u \rangle = 1$ under the Hermitian pairing. We evaluate

$$\begin{aligned} \ell_1 &= \frac{1}{2\omega} \operatorname{Re} \left\langle p, C(u, u, \bar{u}) - 2B\{u, J^{-1}B(u, \bar{u})\} \right. \\ &\quad \left. + B\{\bar{u}, (2i\omega I - J)^{-1}B(u, u)\} \right\rangle. \end{aligned} \quad (17)$$

All derivatives of (4) through order three are analytic. Importantly, the variance equation also contributes $B_{v, v\chi} = B_{v, \chi v} = \gamma$; omitting this term changes ℓ_1 . The repository's symbolic computation and automated tests check the analytic book derivatives, characteristic polynomial, and eigenvalue factorization. A finite-difference reconstruction of B and C provides a separate numerical check on ℓ_1 . The sign of ℓ_1 supplies the local Hopf classification. Its magnitude varies with state and eigenvector normalization, so the analysis uses it as a classification quantity rather than an economic effect size. Sections B and C give the full derivative and computational details.

5 Illustrative nonlinear validation

The configuration in table 1 is a constructive example chosen to produce a well-separated first Hopf crossing with a cycle period of several weeks and a positive-variance orbit. The parameters are illustrative rather than calibrated to SPX.

Object	Value	Interpretation
δ	0.5 yr^{-1}	local reference pull
κ_v, θ_v	$40 \text{ yr}^{-1}, 0.0625 \text{ yr}^{-1}$	variance reversion and level
α, β	$50 \text{ yr}^{-1}, 2$	memory rate and loading
γ	500 yr^{-1}	variance-memory gain
(μ_q, σ_q, T)	$(0.06, 0.20, 0.25 \text{ yr})$	Gaussian signed book
r, q_d	$0, 0$	book-kernel carry inputs
s	$+1$	illustrative latent orientation

Table 1: Illustrative structural parameters. No row is an empirical estimate.

All subsequent computations concern the deterministic drift, so ξ and ρ are not assigned numerical values; they do not enter J , κ^* , ω^* , or ℓ_1 .

At the fixed equilibrium, the normalized book has $G_X = -0.0617978$ and $G_v = -2.0198878$. Enumerating both roots of (14) and applying every side condition gives the first valid point

$$\kappa^* = 31.4932976 \text{ yr}^{-1}, \quad \omega^* = 47.1185670 \text{ rad yr}^{-1}, \quad \ell_1 = -6.2888041. \quad (18)$$

The period $2\pi/\omega^* = 0.1333484$ years is about 33.6 trading days under a 252-day convention, or 48.7 calendar days under ACT/365, and the crossing derivative is $0.2686294 > 0$. Therefore the local normal form predicts stable cycles just above the threshold. The loop gain is $L = 3125$, while the right side of the baseline-stability condition (16) is 368145. At the Hopf point, $(c_2, c_1, c_0) = (92.4462, 2220.1594, 205245.3288)$ and $H'(\kappa^*) = -5784.3769$, so none of the classification conditions is numerically close to zero except the determinant condition being solved.

The sign structure clarifies what drives this example. Although $s = +1$ and $G_X < 0$ strengthen direct price damping, $G_v < 0$ makes the $v \rightarrow X$ link increasingly negative inside the delayed three-state loop. With $\beta, \gamma > 0$, a positive X raises χ and then v ; this price-to-variance sign is opposite the usual equity leverage effect and is an explicit modeling choice. A linear-coefficient ablation that sets $G_v = 0$ has no valid positive Hopf point, whereas setting $G_X = 0$ leaves a crossing at $\kappa = 28.9141 \text{ yr}^{-1}$ and $\omega = 45.2217 \text{ rad yr}^{-1}$. The familiar one-state statement that long-gamma rebalancing is contrarian therefore does not determine stability here: the canonical cycle depends on the book's variance sensitivity and the separately assumed price-memory-variance channel.

The quadratic also has a second valid positive local Hopf point at $\kappa^{**} = 16860.8961 \text{ yr}^{-1}$, with $\omega^{**} = 309.5511 \text{ rad yr}^{-1}$, $d \text{Re } \lambda_+ / d\kappa = -0.0020984$, and $\ell_1 = 17479.5557$. The pair therefore crosses back into the stable half-plane as κ increases through this second root, with an unstable small orbit on the locally stable side. We report it to make the quadratic derivation complete; its coupling is more than 500 times the first threshold, and no nonlinear orbit or economic relevance is claimed in that remote regime.

Figure 1 checks more than the linear algebra. We integrate the actual nonlinear Gaussian-book drift at $1.02\kappa^*$ from a small perturbation, discard 40 years of transient, and plot the final orbit. Over the retained trajectory, $X \in [-0.02814, 0.02358]$, $v \in [0.04453, 0.08187]$, and $\chi \in [-0.04297, 0.03310]$. Thus the displayed cycle remains in the physical variance domain and in a modest log-price neighborhood. The long transient discard is a numerical convergence device rather than an economic forecast horizon.

The magnitude of κ depends on the normalization in (5). It cannot be compared with a per-dollar gamma-exposure coefficient without an explicit unit map. The illustration establishes internal possibility, not empirical proximity.

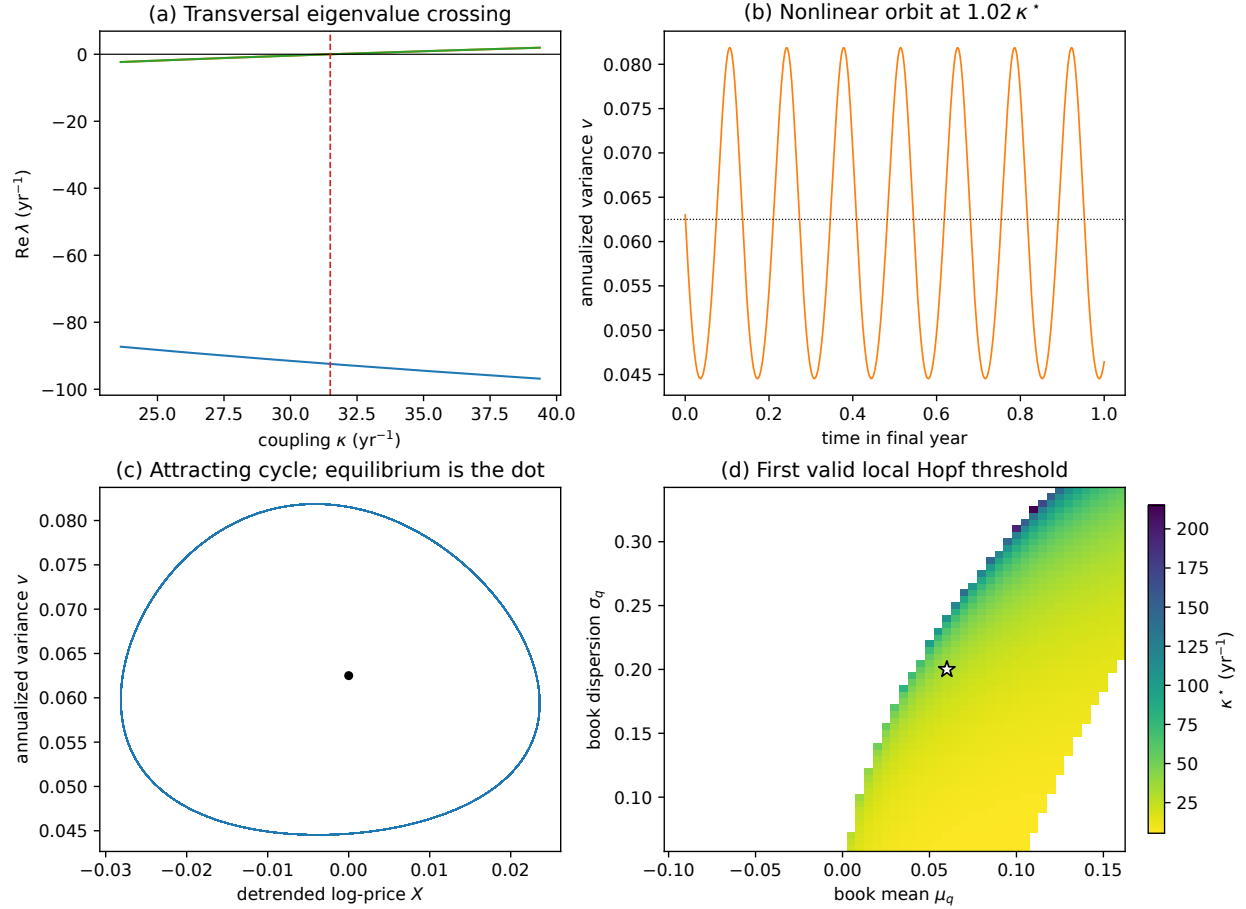


Figure 1: **Fixed-equilibrium Hopf validation using the actual nonlinear book.** (a) Real parts of all three eigenvalues; the complex-conjugate pair overlaps exactly and crosses transversally at the dashed line. (b)–(c) Deterministic trajectory at $1.02\kappa^*$ after the transient. The dotted line in (b) is θ_v and the dot in (c) is the fixed equilibrium. (d) First positive root satisfying all local Hopf side conditions when only (μ_q, σ_q) vary; the star is table 1. Blank cells indicate that no valid local Hopf root was found and carry no global-stability implication. Reproduction command: `python -m reflexive_options.experiments.centered_hopf_validation`.

6 Robustness and numerical validation

A single orbit verifies implementation at one point. Structural stability across book shapes and numerical choices requires a broader sequence of checks on the centered model. These use no market data; the parameter ranges are mechanical perturbations without statistical calibration.

6.1 Local normal-form scaling and numerical convergence

For relative distances $\varepsilon_\kappa = (\kappa - \kappa^*)/\kappa^*$ in $\{0.000625, 0.00125, 0.0025, 0.005, 0.01, 0.02, 0.04, 0.06\}$, we integrate the actual Gaussian-book drift. The transient horizon is at least twelve linear growth time constants, $12/(\operatorname{Re} \lambda'_+(\kappa^*)\kappa^*\varepsilon_\kappa)$, with a floor of 80 years; only the last five simulated years are measured. The unusually long horizon is a convergence device necessitated by critical slowing near the threshold.

On the six points no more than two percent above threshold, a log-log regression of the half peak-to-peak X amplitude on ε_κ has slope 0.5089 and $R^2 = 0.99992$, close to the Hopf prediction $1/2$. Including all points through six percent raises the slope to 0.5338, which documents departure from the leading local normal form rather than contradicting it. The measured period is 0.13341 years at 0.0625% above threshold, compared with the linear prediction 0.13335; it lengthens to 0.15667 by six percent. Thus both amplitude and period behave locally as the theorem predicts, while finite-distance corrections become visible.

At two percent above threshold, we repeat the calculation with the explicit eighth-order DOP853 integrator after tightening its relative and absolute error tolerances according to

$$(\text{rtol}, \text{atol}) : (10^{-8}, 10^{-10}) \longrightarrow (10^{-10}, 10^{-12}).$$

This changes the X amplitude by 1.94×10^{-11} relatively and does not change the reported period at the measurement resolution. Four initial conditions spanning both signs of X , variance offsets of ± 0.003 , and a non-zero memory perturbation converge to amplitudes with coefficient of variation 3.12×10^{-8} and periods with coefficient of variation 3.65×10^{-6} . These are numerical statements about the displayed attractor, not a proof of a global basin.

Panel (c) shows that the first threshold changes nonlinearly under one-at-a-time parameter perturbations and can disappear when a local side condition fails. The sweep is a diagnostic of model dependence, not a statistical confidence region.

6.2 Dependence on book shape and mixtures

On a 45×45 equally spaced grid with $\mu_q \in [-0.10, 0.16]$ and $\sigma_q \in [0.06, 0.34]$, 747 of the 2,025 cells have a valid positive local Hopf root; 722 are supercritical and 25 subcritical. The grid varies only two shape parameters with all dynamical parameters fixed, so its proportions are not probabilities. The blank regions show strong model dependence: the local Hopf mechanism requires an appropriate shape and signed orientation of the latent book.

To move beyond a single Gaussian, let component j have positive mass w_j , orientation $s_j \in \{-1, +1\}$, and raw kernel $\mathcal{G}_j$. We use

$$g_{\text{mix}}(X, v) = \frac{\sum_j w_j s_j \{\mathcal{G}_j(X, v) - \mathcal{G}_j(0, \theta_v)\}}{\sum_j w_j \mathcal{G}_j(0, \theta_v)}. \quad (19)$$

The denominator is gross equilibrium gamma. A signed-net denominator can become arbitrarily small when long and short components nearly cancel, producing an artificial divergence in the

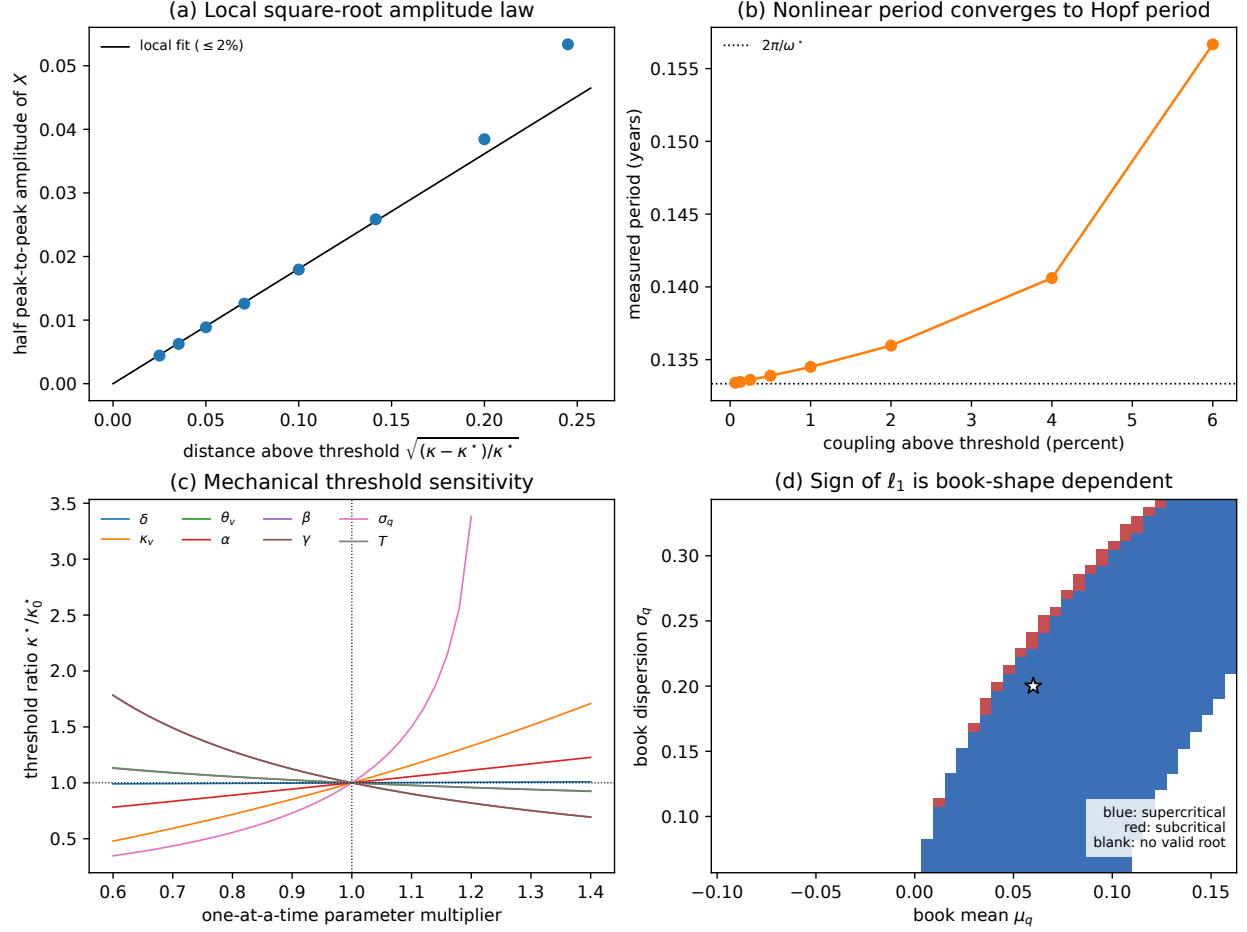


Figure 2: **Centered-model robustness.** (a) The closest nonlinear cycles follow the square-root amplitude law. (b) Their periods approach $2\pi/\omega^*$ at the threshold. (c) One-at-a-time perturbations from 60% to 140% of each canonical value show mechanical, not statistical, threshold sensitivity; gaps denote failure of a local Hopf side condition. (d) Over a 45×45 (μ_q, σ_q) grid, blue cells are supercritical, red cells subcritical, and blank cells have no positive root satisfying all local side conditions. Reproduction command: `python -m reflexive_options.experiments.centered_model_robustness`.

normalized coupling. Because differentiation is linear, all derivatives in eq. (17) are gross-normalized weighted sums of the component derivatives.

The diagnostic mixtures are fully specified as $w \times (\mu_q, \sigma_q, T, s)$ components. The nearby book is $0.5 \times (0.04, 0.18, 0.20, +1) + 0.5 \times (0.08, 0.22, 0.30, +1)$; the dispersed book is $0.6 \times (0.02, 0.12, 0.10, +1) + 0.4 \times (0.10, 0.24, 0.50, +1)$; and the offsetting book uses the same two component shapes with weights 0.7 and 0.3 but orientation -1 on the second component. These choices probe nearby, maturity-dispersed, and sign-offsetting books rather than representing estimated portfolios.

Book	G_X	G_v	κ^*	ℓ_1	Local classification
Canonical single	-0.0618	-2.0199	31.4933	-6.2888	supercritical
Nearby same-sign mixture	-0.1152	-2.0132	34.3891	-4.8143	supercritical
Dispersed same-sign mixture	-0.1514	-2.3480	30.2416	5.8498	subcritical
Offsetting-sign mixture	-0.1440	-1.5645	50.4759	27.8170	subcritical
Reversed canonical orientation	0.0618	2.0199	—	—	no valid positive root

Table 2: Gross-normalized mixture analysis. The examples are diagnostic constructions, not estimated books.

Table 2 shows that the existence of a valid Hopf root survives several deformations, whereas supercriticality does not: second and third derivatives of the book can flip ℓ_1 without eliminating the imaginary-axis crossing. A subcritical result establishes only a local unstable orbit on the opposite side; it does not establish hysteresis or a large stable cycle. Reversing the canonical orientation produces no positive root satisfying all side conditions. Therefore neither the existence nor the type of the local bifurcation can be inferred from unsigned open interest.

6.3 Independent derivative check

Finally, we reconstruct every entry of B and C by central finite differences of the full drift at step 2×10^{-4} , without using the analytic book derivatives. The result is $\ell_1 = -6.288803983$, compared with -6.288804057 analytically: absolute difference 7.41×10^{-8} and relative difference 1.18×10^{-8} . This check is especially important because omitting the mixed derivative of $\gamma v \chi$ can preserve the Hopf threshold while changing its nonlinear classification.

7 Empirical estimand and pre-extraction protocol

The empirical exercise targets the linear-projection coefficient of next-session log squared return on one standardized observable book summary, conditional on the fixed controls below. Its interpretation is incremental predictive association. The structural coupling κ , the book derivatives G_X and G_v , signed dealer inventory, and causal responses lie outside this regression’s estimand.

The connection to the model is deliberately limited but substantive. The theorem makes the level and derivatives of latent book geometry relevant to local dynamics, while public data preserve only unsigned mass, call-put composition, and moneyness shape. The protocol therefore asks whether these observable projections carry incremental predictive information for a next-session variance proxy. Association would motivate a later structural study with participant-tagged positions; this design imposes no directional prediction from the Hopf example.

No OptionMetrics option-chain data, Center for Research in Security Prices (CRSP) returns, or Cboe Volatility Index (VIX) data have been loaded or analyzed for this paper. Because the historical outcomes and named stress episodes were public when the protocol was written, the design

is a retrospective pre-analysis plan recorded before proprietary-data extraction. In the repository’s cumulative registration ledger, Amendments A13–A16 replace the earlier event-selected directional gamma-exposure plan and then correct its calendar, availability, and measurement rules; A1–A12 document superseded designs. The timestamp and file-hash record appears in section D.

7.1 Primary proxy construction

On each eligible OptionMetrics date from 3 January 2017 through 29 October 2024, retain contracts whose verified Options Symbology Initiative root is SPX or SPXW. The latter includes the weekly and daily-expiration series. Keep only unadjusted, standard-settlement contracts with multiplier 100, a settlement timestamp strictly after the valuation close, $0 < \tau \leq 1$ under fractional ACT/365 time, implied volatility (IV) in $[0.01, 5]$, and $|\log(K/F)| \leq 0.50$. The primary quote filter requires positive bid, ask above bid, and relative spread no greater than 0.50.

Amendment A16 uses the latest open-interest record documented as available by 16:00 America/New_York on regressor session t . If same-date availability is not documented, the previous available record is used; ambiguity defaults to the previous CRSP session. The chosen availability branch and vendor documentation are frozen before returns are extracted.

Form the parity forward separately within each date–expiration–AM/PM-settlement group. Let C_{mid} and P_{mid} be the matched call and put midquotes, let τ be fractional ACT/365 time to settlement, and let r be the continuously compounded zero rate. Choose the strike minimizing $|C_{\text{mid}} - P_{\text{mid}}|$ and set

$$F = K + e^{r\tau}(C_{\text{mid}} - P_{\text{mid}}).$$

When no matched pair survives, use the flagged carry fallback $F = Se^{(r-q_d)\tau}$. Here S is the OptionMetrics SPX close, q_d is the index-dividend yield, and r is linearly interpolated from the continuous zero curve. After these fixed filters, construct the nonnegative gamma mass

$$U_t = \sum_i \text{OI}_{i,t}^{\text{avail}} \Gamma_{i,t}^{BS} S_t^2(0.01)(100), \quad (20)$$

where the factors 0.01 and 100 express gamma exposure for a one-percent underlying move and the standard contract multiplier. Together with U_t , construct three observable shape summaries. Write $u_{i,t} = \text{OI}_{i,t}^{\text{avail}} \Gamma_{i,t}^{BS} S_t^2(0.01)(100)$ and $k_{i,t} = \log(K_i/F_{i,t})$, and let C and P index calls and puts. The summaries are

$$B_t = \frac{\sum_{i \in C} u_{i,t} - \sum_{i \in P} u_{i,t}}{U_t}, \quad M_t = \frac{\sum_i u_{i,t} k_{i,t}}{U_t}, \quad D_t = \left\{ \frac{\sum_i u_{i,t} (k_{i,t} - M_t)^2}{U_t} \right\}^{1/2}. \quad (21)$$

These are, respectively, call–put composition, gamma-weighted mean moneyness, and dispersion. They are participant-sign-free, OI-and-pricing-derived summaries: their weights also depend on spot, IV, rates, dividends, maturity, and forward moneyness. They are not estimates of dealer holdings or pure position measures. Convention-based signed gamma-exposure series, including the original SqueezeMetrics construction (SqueezeMetrics, 2017), are reported only as secondary measurement sensitivities. Reversing a convention mechanically reverses its coefficient and is not robustness evidence.

The primary outcome is $Y_{t+1} = \log(r_{t+1}^2 + 10^{-8})$, where 10^{-8} is one basis point squared when returns are expressed as decimals, and the immediately following CRSP trading-session log return is $r_{t+1} = \log(1 + \text{crsp.dsp500.sprtrn}_{t+1})$. The VIX control is the archived official Cboe daily close.

For each standardized proxy, estimate a strictly forward regression

$$\begin{aligned}
Y_{t+1} = & a + bQ_t + \phi_1 Y_t + \phi_5 \bar{Y}_{t-4:t} + \phi_{22} \bar{Y}_{t-21:t} \\
& + \eta \log(\text{VIX}_t) + \zeta \log S_t + \psi L_t \\
& + \text{outcome-weekday and monthly-expiration controls} + e_{t+1}.
\end{aligned} \tag{22}$$

Here Q_t is, in separate prespecified regressions, $\log U_t$, B_t , M_t , or $\log(D_t + 10^{-6})$, and L_t is a centered linear CRSP-session trend. The bars denote five-session and 22-session trailing means, $\bar{Y}_{t-4:t} = 5^{-1} \sum_{j=0}^4 Y_{t-j}$ and $\bar{Y}_{t-21:t} = 22^{-1} \sum_{j=0}^{21} Y_{t-j}$. Leads and lags are constructed on the complete ordered CRSP trading-session calendar before rows with a missing option proxy are removed. Thus a missing option date cannot compress the return horizon: $t + 1$ always denotes the immediately following CRSP session. Tuesday–Friday indicators refer to that outcome session, whose calendar identity is known at the close of t ; the monthly-expiration indicator refers to the regressor session. The primary inferential target is incremental predictive association, $b = 0$ against a two-sided alternative. Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) inference uses Bartlett weights at lag 22 measured in actual CRSP-session offsets (Newey and West, 1987). The moving-pairs bootstrap uses length-22 blocks on the complete session index, 2,000 full-rank draws, seed 42, and a 95% percentile interval (Künsch, 1989). Missing option rows therefore compress neither return horizons nor inference lags; the former lag-5 and retained-row block-10 procedures are sensitivities.

Benjamini–Hochberg false-discovery-rate (FDR) control at $q_{\text{FDR}} = 0.05$ is applied separately to the four HAC and four bootstrap p -values (Benjamini and Hochberg, 1995). A proxy is labeled robustly associated only when both adjusted p -values are below 0.05 and the bootstrap interval excludes zero; a one-family rejection is labeled method-sensitive, and every other case is labeled not detected. The common coefficient and fixed decision rule prevent ex-post selection between inference methods.

Object	Locked source	Role and principal limitation
Option chain, OI, IV, spot	OptionMetrics IvyDB US	OI-and-pricing summaries; OI lacks participant identity and position side
Forward, rates, dividends	Put–call parity by settlement group; OptionMetrics zero curve and index dividend	Fractional settlement time; carry is a flagged fallback
S&P 500 return	<code>crsp.dsp500.sprtrn</code>	Uses the vendor return for the next CRSP session
VIX	Archived official Cboe daily close	Predetermined close- t control from one source
Expiration calendar	Archived official Cboe expiration and holiday calendar	Standard monthly SPX third Friday, shifted to the preceding open session if closed

Table 3: Prespecified source map. Raw files, query text, retrieval times, row counts, date coverage, and SHA-256 hashes are frozen before outcomes are constructed.

The repository currently provides tested construction and inference utilities rather than a complete Wharton Research Data Services (WRDS) extraction pipeline. A16 therefore requires a production adapter that halts on an unresolved required field, timing rule, or duplicate, a sequential attrition log, availability-timing gate, and frozen raw manifests before outcomes are joined. The empirical code is not represented as operationally complete until that gate passes.

The overlapping Y_t , five-day, and 22-day controls are intentionally fixed as a heterogeneous-autoregressive (HAR)-style predictive conditioning set. Their collinearity can inflate uncertainty but does not by itself create ordinary-least-squares (OLS) bias. The primary table reports variance-inflation factors (VIFs) rather than changing controls after seeing results. Proxy standardization uses one common option-date sample before outcome-based listwise deletion. Because that transformation uses no future return and is affine, it changes coefficient units but not fitted values or the test of a zero coefficient.

The bootstrap resamples complete-calendar blocks of aligned outcome-and-design rows and uses the same random starts in all four regressions. Rank-deficient draws are replaced and counted. The reported bootstrap p -value is the Monte Carlo-corrected two-sided sign-tail probability, and the interval is percentile-based. HAC and bootstrap are two uncertainty approximations for one coefficient rather than independent replications. The conjunctive label is a prespecified robustness convention; raw and adjusted values and the full proxy correlation matrix will be reported because the formal FDR guarantee depends on conditions that may fail for correlated summaries.

A16 also fixes measurement decompositions before extraction. The report includes year effects, first differences, a mass normalized by total OI, three subperiod estimates, OI-only shape summaries, and gamma weights recomputed at fixed 20% volatility. These diagnostics separate persistence and contemporaneous option-surface inputs from changes in the OI cross-section; they cannot replace the primary specification.

Event-window interactions for Volmageddon, COVID, and the August 2024 yen-carry episode are secondary heterogeneity analyses, and absolute return is a secondary outcome. Neither affects sample construction or the primary decision rule.

7.2 Ex-ante precision calculation

Detectable effect size depends on sample size, serial dependence, and the extent to which each proxy is explained by the controls. To expose that scale before extraction, residualize a standardized proxy Q on the controls and let R_Q^2 be the auxiliary regression's R^2 . Under an idealized Gaussian scenario with effective independent sample size n_{eff} , residual outcome standard deviation σ_e , two-sided five-percent size, and 80-percent power, let z_p denote the p quantile of the standard normal distribution. The minimum detectable coefficient is approximately

$$\frac{|b|_{\text{MDE}}}{\sigma_e} \simeq \frac{z_{0.975} + z_{0.80}}{\sqrt{n_{\text{eff}}(1 - R_Q^2)}} = \frac{2.802}{\sqrt{n_{\text{eff}}(1 - R_Q^2)}}. \quad (23)$$

n_{eff}	$R_Q^2 = 0$	$R_Q^2 = 0.5$
500	$0.125 \sigma_e$	$0.177 \sigma_e$
1000	$0.089 \sigma_e$	$0.125 \sigma_e$
1500	$0.072 \sigma_e$	$0.102 \sigma_e$
2000	$0.063 \sigma_e$	$0.089 \sigma_e$

Table 4: Idealized 80-percent-power sensitivity from eq. (23). Serial dependence, heavy tails, missing rows, collinearity, and multiplicity reduce effective precision; the table is not a claim about realized power.

No realized n_{eff} estimator is prespecified; the table is a scenario calculation. The empirical report will give raw n , R_Q^2 , σ_e , and the HAC/bootstrap uncertainty directly. Because the design

specifies no smallest effect of interest, a null result supports “not detected under this design” but cannot establish practical equivalence to zero.

7.3 Decision language fixed before extraction

The following interpretation is part of the design.

- If no registered book summary has an incremental association, predictive relevance at the daily horizon is not detected under these measurements. Without an equivalence margin, failure to reject is not evidence that every economically meaningful effect is absent.
- An association for U_t , B_t , M_t , or D_t supports predictive content in an OI-and-pricing-derived book summary, but does not identify dealer sign or the structural feedback in (1).
- Results that depend on a sign convention are labeled non-identified. They cannot be reported as confirmation that dealers were long or short gamma.
- Even a robust predictive association is not causal. A direct test of the signed-book mechanism requires participant-tagged positions or another independently validated dealer- inventory measure, plus an identification strategy for endogenous option demand.

These rules fix the interpretation before the coefficient signs are observed.

8 Limitations and falsifiable scope

The analysis has substantial limitations.

1. **Local deterministic result.** The theorem characterizes the drift near one equilibrium. Global attraction, global non-explosion, an ergodic distribution, tail behavior, and stochastic bifurcation under state-dependent noise require separate analyses (Baxendale, 1994, 2025).
2. **Coordinate dependence.** Deleting diffusion from an Itô equation is not invariant under nonlinear state transformations. The reported Hopf point belongs to the chosen detrended-log-price drift; it is not a coordinate-free stochastic threshold.
3. **Reduced-form mechanism.** The centered gamma level is a shape restriction on price pressure rather than hedge flow or an impact-cleared demand equation. The assumed positive price–memory–variance channel is indispensable, has the opposite sign from the usual equity leverage effect, and makes the canonical crossing depend mainly on G_v .
4. **Frozen low-dimensional book.** Actual positions evolve across strike, maturity, and participant. The Gaussian kernel, common variance input, and frozen effective maturity suppress the implied-volatility surface, inventory replenishment, and the variance-risk-premium wedge. The mixture analysis confirms that nonlinear classification depends on book shape.
5. **Measurement and theory–proxy gap.** Public OI does not recover signed dealer inventory. The empirical summaries also contain contemporaneous pricing inputs. Their predictive content cannot directly test the theorem’s signed channel or isolate position changes without the A16 decompositions.

6. **Predictive implementation and inference.** Log squared return is noisy, the controls may be collinear, and HAC, calendar-block bootstrap, and FDR calculations rely on approximations. The production WRDS adapter, attrition ledger, and timing gate must still be completed before outcomes are joined.
7. **Retrospective end-of-day horizon.** The 2017–2024 outcomes were historically observable when the plan was written. The next-session design excludes contracts settled by the regressor close and therefore cannot test intraday zero-days-to-expiration feedback.
8. **Causal identification.** Option demand, inventory, volatility expectations, and returns are jointly determined. Forward dating prevents mechanical look-ahead but does not remove omitted variables, anticipation, measurement error, or equilibrium endogeneity.

The supported claim is consequently precise: the specified local reduced-form system can undergo a non-degenerate Hopf bifurcation under explicit conditions, and one transparent nonlinear example satisfies them. Whether dealer positions generate a related mechanism in market data remains open.

9 Reproducibility and conclusion

The source, analytic derivatives, numerical experiment, and tests are available at <https://github.com/mahimn01/reflexive-options>. The automated and cross-method checks verify that the equilibrium does not move with κ , the variance-boundary drift is inward for arbitrary memory values, the quadratic (14) matches the direct characteristic polynomial, the stated eigenvalues occur at the reported root, and the nonlinear orbit keeps $v > 0$. The arXiv bundle contains only referenced assets and is built from a clean extraction test.

The theoretical result is a local mechanism: consistent treatment of the common equilibrium, variance boundary, nonlinear derivatives, and Routh–Hurwitz conditions produces an endogenous volatility cycle in the specified three-state system. The empirical boundary is equally clear: public open interest is not dealer inventory. The preregistered exercise therefore reports the predictive content of OI-and-pricing-derived book summaries and limits its interpretation to the estimand the data support.

A Derivation of the Gaussian book integral

Normalize the reference forward $F_\star$ to one, write $S = e^X$, and let $k = \log K$. Black–Scholes spot gamma for a common maturity T is

$$\Gamma^{BS}(X, k, v, T) = \frac{e^{-q_d T - X}}{\sqrt{vT}} \phi\left(\frac{X - k + (r - q_d + v/2)T}{\sqrt{vT}}\right), \quad (24)$$

where ϕ is the standard-normal density. Write $\varphi(k; \mu, \sigma^2)$ for a Gaussian density with mean μ and variance σ^2 . With $k \sim N(\mu_q, \sigma_q^2)$, the positive unit-mass book is

$$\begin{aligned} \mathcal{G}(X, v) &= \int_{\mathbb{R}} \varphi_q(k) \Gamma^{BS}(X, k, v, T) dk \\ &= e^{-q_d T - X} \int_{\mathbb{R}} \varphi(k; \mu_q, \sigma_q^2) \varphi\{k; X + (r - q_d + v/2)T, vT\} dk. \end{aligned} \quad (25)$$

The integral of two Gaussian densities in the same variable is the Gaussian density of the difference of their means with variance equal to the sum:

$$\int \varphi(k; \mu_1, s_1^2) \varphi(k; \mu_2, s_2^2) dk = \varphi(\mu_1; \mu_2, s_1^2 + s_2^2).$$

Applying the identity to eq. (25), setting $\tau^2(v) = \sigma_q^2 + vT$ and $m(v) = \mu_q - (r - q_d + v/2)T$, gives eq. (4). A direct completion of the square yields the same expression. Because $\sigma_q > 0$, the convolved formula is smooth in v on a neighborhood of every $\theta_v > 0$.

The strike distribution remains fixed as spot moves, so the dependence on X arises through changing moneyness. Proposition 2 absorbs gross contract mass, contract multiplier, and a common impact coefficient into the normalization of κ .

B Analytic derivative details

Let $D = X - m(v)$, $t = \tau^2(v)$, and $h = \log \mathcal{G}$. Constants independent of (X, v) can be suppressed, leaving

$$h = -X - \frac{1}{2} \log t - \frac{D^2}{2t}, \quad D_v = T/2, \quad t_v = T.$$

The log-derivatives needed through order three are

$$\begin{aligned} h_X &= -1 - D/t, & h_{XX} &= -1/t, & h_{XXX} &= 0, \\ h_v &= -\frac{T}{2t} - \frac{DT}{2t} + \frac{D^2 T}{2t^2}, \\ h_{vv} &= \frac{T^2}{2t^2} - \frac{T^2}{4t} + \frac{DT^2}{t^2} - \frac{D^2 T^2}{t^3}, \\ h_{vvv} &= -\frac{T^3}{t^3} + \frac{3T^3}{4t^2} - \frac{3DT^3}{t^3} + \frac{3D^2 T^3}{t^4}, \\ h_{Xv} &= -\frac{T}{2t} + \frac{DT}{t^2}, & h_{XXv} &= \frac{T}{t^2}, & h_{Xvv} &= \frac{T^2}{t^2} - \frac{2DT^2}{t^3}. \end{aligned} \tag{26}$$

For indices $i, j, k \in \{X, v\}$, raw derivatives follow from

$$\begin{aligned} \mathcal{G}_i &= \mathcal{G} h_i, \\ \mathcal{G}_{ij} &= \mathcal{G} (h_i h_j + h_{ij}), \\ \mathcal{G}_{ijk} &= \mathcal{G} (h_i h_j h_k + h_i h_{jk} + h_j h_{ik} + h_k h_{ij} + h_{ijk}). \end{aligned} \tag{27}$$

The centered constant in eq. (5) has zero derivative, so every non-zero derivative of g is $s\mathcal{G}_{i...}/\mathcal{G}(0, \theta_v)$.

At the canonical configuration, the complete non-zero book derivative set used in B and C is

Derivative	Value	Derivative	Value	Derivative	Value
G_X	-0.0617978	G_v	-2.0198878	G_{XX}	-17.9737091
G_{Xv}	-6.3390058	G_{vv}	11.8556078	G_{XXX}	3.3326765
G_{XXv}	117.9016552	G_{Xvv}	83.4816197	G_{vvv}	-114.8072302

Table 5: Analytic normalized derivatives at $(0, \theta_v)$. Symbolic differentiation of eq. (4) agrees to displayed precision.

For the canonical drift expansion at the first Hopf point, $f(z_* + x; \kappa^*) = J(\kappa^*)x + \frac{1}{2}B(x, x) + \frac{1}{6}C(x, x, x) + O(\|x\|^4)$, the book contributes

$$\begin{aligned} B_{1,XX} &= \kappa^* G_{XX}, & B_{1,Xv} &= B_{1,vX} = \kappa^* G_{Xv}, & B_{1,vv} &= \kappa^* G_{vv}, \\ C_{1,XXX} &= \kappa^* G_{XXX}, & C_{1,XXv} &= \kappa^* G_{XXv}, & C_{1,Xvv} &= \kappa^* G_{Xvv}, & C_{1,vvv} &= \kappa^* G_{vvv}, \end{aligned}$$

with all permutations of the last three indices filled symmetrically. The variance drift adds $B_{2,v\chi} = B_{2,\chi v} = \gamma$. There are no other quadratic or cubic drift derivatives in the static Gaussian specialization.

C Algebraic and numerical details

For completeness, with $G_\chi = 0$ the coefficients are

$$\begin{aligned} c_2 &= A + \delta - \kappa G_X, \\ c_1 &= M + \delta A - \kappa G_X A, \\ c_0 &= \delta M + \frac{1}{2}L - \kappa(G_X M + G_v L). \end{aligned}$$

Multiplying the first two and subtracting the third gives

$$\begin{aligned} H(\kappa) &= G_X^2 A \kappa^2 + \{G_v L - G_X A(A + 2\delta)\} \kappa \\ &\quad + A(M + \delta A + \delta^2) - \frac{1}{2}L, \end{aligned}$$

which is (14)–(15). If $G_X = 0$, the equation is linear unless $G_v L = 0$; the implementation handles this degeneracy separately. If the discriminant is non-negative, both real roots are enumerated in ascending order and each is subjected to the positivity, frequency, factorization, and transversality checks. Thus the reported root is not selected by sign alone.

At a valid root, factorization gives $P_\lambda(i\omega) = 2\{-c_1 + ic_2\sqrt{c_1}\}$. Combining this with $P_\kappa(i\omega)$, using $c_0 = c_1 c_2$, and taking the real part of $-P_\kappa/P_\lambda$ yields (13). This also shows why $H' \neq 0$ is the relevant crossing condition but does not, by itself, determine which side contains stable cycles; that requires the sign of both (13) and ℓ_1 .

C.1 Root-classification algorithm

The implementation follows the following deterministic sequence.

1. Compute analytic G_X, G_v and (A_2, A_1, A_0) .
2. Treat the polynomial as linear exactly when $A_2 = 0$. Otherwise compute the quadratic discriminant and compare a negative value with the relative tolerance $10^{-8} \max(A_1^2, |4A_2 A_0|)$ before enumerating both real roots with the cancellation-resistant formula.
3. Sort positive roots. For each root, rebuild J directly and compute the characteristic polynomial numerically rather than reusing the quadratic coefficients.
4. Require $c_2, c_1, c_0 > 0$, a scaled residual $|c_1 c_2 - c_0| \leq 10^{-7} \max(1, |c_0|, |c_1 c_2|)$, non-zero frequency, eigenvalue factorization within $10^{-6} \max(1, \omega)$, and transversality whose H' magnitude exceeds $10^{-8} \max(|2A_2 \kappa|, |A_1|)$.

5. Construct analytic B, C and evaluate eq. (17) at every root passing the linear checks. Return all valid points in ascending order. Analyses explicitly labeled “threshold” use the first point; otherwise all points are reported. If none passes, report that no valid positive local Hopf root was found.

The algorithm never converts a negative discriminant into a stability conclusion.

C.2 Nonlinear convergence table

$100\varepsilon_\kappa$	Transient horizon	A_X	A_v	Period	v range
0.0625	2269.50	0.00442	0.00325	0.13341	[0.05927, 0.06577]
0.125	1134.75	0.00626	0.00460	0.13347	[0.05795, 0.06715]
0.25	567.37	0.00887	0.00651	0.13361	[0.05608, 0.06910]
0.5	283.69	0.01259	0.00922	0.13389	[0.05346, 0.07191]
1.0	141.84	0.01795	0.01309	0.13450	[0.04978, 0.07595]
2.0	80.00	0.02586	0.01867	0.13596	[0.04453, 0.08187]
4.0	80.00	0.03844	0.02713	0.14061	[0.03664, 0.09089]
6.0	80.00	0.05337	0.03625	0.15667	[0.02769, 0.10018]

Table 6: Actual-book deterministic cycles. A_X and A_v are half peak-to-peak amplitudes measured over the final five simulated years. All times are in years.

The solver is DOP853 with stated tolerances ($10^{-9}, 10^{-11}$) and maximum step 0.005 years. Periods are averages of successive local maxima of X over the retained window. The plotted $1.02\kappa^*$ orbit in fig. 1 is the same configuration as the 2.0-percent row of table 6; the earlier 40-year discard is more than sufficient for the plotted initialization, while the robustness validation uses the more conservative 80-year rule.

D Empirical implementation and reporting plan

The data workflow is ordered to prevent outcome knowledge from influencing contract construction.

The timestamped original registration and its amendments remain in the repository. Amendments A13–A14 close the contract, source, and inference choices before extraction; A15 fixes market-calendar alignment, the outcome-session weekday control, and the retrospective-registration language. A16 adds information-availability timing, fractional settlement time, liquidity filters, complete-calendar inference, persistence controls, and pricing-input decompositions. The A13–A14 file has SHA-256

603e89366c0dbe49718e8c31f805d6f85d3c508e2e0ed4276a6310c80f5f9cd7

and receipt `pre_registration_amendments.md.ots`. A15 has SHA-256

a5f694f99953d57563d4f17dc5646ef0b87452c45119ccbd12fc90efd034a52

and receipt `pre_registration_amendment_a15.md.ots`. A16 has SHA-256

a5cbf9ef56c9a402ff05b61bb720d8487313b154f30393b4835b43fe5c33e61d

and receipt `pre_registration_amendment_a16.md.ots`. The receipts were pending Bitcoin consolidation when checked on 12 July 2026; earlier records are preserved rather than overwritten.

1. Archive vendor documentation, query text, raw extracts, retrieval time, row count, date range, and SHA-256 hashes. Do not construct returns at this stage.
2. Verify SPX security identifiers, OSI roots, adjustment and settlement fields, contract multipliers, unique option identifiers, the OptionMetrics spot mapping, and OI availability semantics. Non-identical duplicates halt the pipeline and require a documented mapping resolution before analysis continues.
3. Apply every contract filter in A14 and A16 and record sequential attrition: input rows, root/security, adjustment/multiplier, settlement timestamp, quote liquidity, IV, duplicate, forward, and moneyness stages. Record zero-OI rows even if later dropped for speed.
4. Construct parity forwards within date–expiration–settlement groups. Record the fraction of gamma mass using carry fallback and zero-curve endpoint extension. Compare recomputed gamma with vendor gamma only as a mapping diagnostic.
5. Choose the A16 OI-availability branch from archived vendor documentation, construct (U, B, M, D) with fractional settlement time, and freeze the common finite option-date sample. No daily summary is winsorized, clipped, or removed because it is extreme.
6. Only after the option-side freeze, archive CRSP returns and official Cboe VIX. Build the complete ordered CRSP trading-session calendar, construct return leads and lags before removing any date with a missing option proxy, use outcome-session weekday indicators, and record every listwise deletion as fixed in A15–A16.
7. Run all four primary equations with the same controls, including log spot and the linear session trend, and aligned rows. Save the design-matrix hashes, coefficients, HAC covariance matrices, bootstrap seeds and block indices, percentile distributions, VIFs, proxy correlations, and adjusted p -values.
8. Produce the full primary table before the prespecified measurement decompositions, event interactions, or convention-signed sensitivities are opened.

For transparency, the Benjamini–Hochberg adjusted value for ordered raw values $p_{(1)} \leq \dots \leq p_{(m)}$ is

$$\tilde{p}_{(i)} = \min_{j \geq i} \min\{1, mp_{(j)}/j\},$$

mapped back to the original proxy order. HAC and bootstrap families are adjusted separately as locked in A14. Under A16, the moving-block bootstrap draws starting sessions uniformly from all overlapping length-22 blocks on the complete CRSP calendar, concatenates retained rows to obtain n , replaces and counts rank-deficient draws, and truncates the last block. The same starts are used for every primary proxy.

Regardless of the results, the final empirical report will include the complete attrition table; excluded dates and reasons; parity-fallback mass; descriptive distributions in raw units; the four-by-four proxy correlation matrix; VIFs; coefficient, HAC standard error and raw/adjusted p -value; bootstrap standard error, raw/adjusted p -value and interval; prespecified label; availability lag; liquidity and measurement decompositions; all signed-convention sensitivities; and a deviations table comparing every post-data change with the registered analysis. Participant-level or causal interpretations require new data and a new identification design that supports those claims.

References

- Benjamin Andreegg, Florian Ulmann, and Didier Sornette. The impact of option hedging on the spot market volatility. *Journal of International Money and Finance*, 124:102627, 2022. doi: 10.1016/j.jimonfin.2022.102627.
- Guido Baltussen, Zhi Da, Sten Lammers, and Martin Martens. Hedging demand and market intraday momentum. *Journal of Financial Economics*, 142(1):377–403, 2021. doi: 10.1016/j.jfineco.2021.04.029.
- Andrea Barbon and Andrea Buraschi. Gamma fragility. Technical Report 2020/05, University of St. Gallen School of Finance, 2020.
- Peter H. Baxendale. A stochastic Hopf bifurcation. *Probability Theory and Related Fields*, 99:581–616, 1994. doi: 10.1007/BF01206233.
- Peter H. Baxendale. Lyapunov exponents and shear-induced chaos for a Hopf bifurcation with additive noise. *Probability Theory and Related Fields*, 193:375–425, 2025. ISSN 0178-8051. doi: 10.1007/s00440-024-01301-4. Published online 26 July 2024.
- Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.
- William A. Brock and Cars H. Hommes. Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control*, 22(8–9):1235–1274, 1998. doi: 10.1016/S0165-1889(98)00011-6.
- William A. Brock, Cars H. Hommes, and Florian O. O. Wagener. More hedging instruments may destabilize markets. *Journal of Economic Dynamics and Control*, 33(11):1912–1928, 2009. doi: 10.1016/j.jedc.2009.05.004.
- Boyd Buis, Mary Pieterse-Bloem, Willem F. C. Verschoor, and Remco C. J. Zwinkels. Gamma positioning and market quality. *Journal of Economic Dynamics and Control*, 164:104880, 2024. doi: 10.1016/j.jedc.2024.104880.
- Chukwuma Dim, Bjørn Eraker, and Grigory Vilkov. 0DTEs: Trading, gamma risk, and volatility propagation. *SSRN Working Paper*, 2024. doi: 10.2139/ssrn.4692190.
- William Feller. Two singular diffusion problems. *Annals of Mathematics*, 54(1):173–182, 1951.
- Rüdiger Frey and Alexander Stremme. Market volatility and feedback effects from dynamic hedging. *Mathematical Finance*, 7(4):351–374, 1997. doi: 10.1111/1467-9965.00036.
- Nicolae Gârleanu, Lasse Heje Pedersen, and Allen M. Poteshman. Demand-based option pricing. *Review of Financial Studies*, 22(10):4259–4299, 2009. doi: 10.1093/rfs/hhp005.
- Xue-Zhong He and Min Zheng. The dynamics of moving average rules in a continuous-time financial market model. *Journal of Economic Behavior & Organization*, 76(3):615–634, 2010. doi: 10.1016/j.jebo.2010.08.005.
- Xue-Zhong He, Kai Li, Junjie Wei, and Min Zheng. Market stability switches in a continuous-time financial market with heterogeneous beliefs. *Economic Modelling*, 26(6):1432–1442, 2009. doi: 10.1016/j.econmod.2009.07.016.
- Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343, 1993. doi: 10.1093/rfs/6.2.327.
- Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989. doi: 10.1214/aos/1176347265.
- Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Springer, New York, 3rd edition, 2004. doi: 10.1007/978-1-4757-3978-7.
- W.-M. Liu. Criterion of Hopf bifurcations without using eigenvalues. *Journal of Mathematical Analysis and Applications*, 182(1):250–256, 1994. doi: 10.1006/jmaa.1994.1079.
- Roger Lord, Remmert Koekoek, and Dick van Dijk. A comparison of biased simulation schemes for stochastic volatility models. *Quantitative Finance*, 10(2):177–194, 2010. doi: 10.1080/14697680802392496.

- Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987. doi: 10.2307/1913610.
- Options Industry Council. Open interest: Why it matters. Options Education, October 2025. URL <https://www.optionseducation.org/news/open-interest-why-it-matters>. Accessed 9 July 2026.
- Eckhard Platen and Martin Schweizer. On feedback effects from hedging derivatives. *Mathematical Finance*, 8(1):67–84, 1998. doi: 10.1111/1467-9965.00045.
- K. Ronnie Sircar and George Papanicolaou. General Black–Scholes models accounting for increased market volatility from hedging strategies. *Applied Mathematical Finance*, 5(1):45–82, 1998. doi: 10.1080/135048698334727.
- SqueezeMetrics. The implied order book and gamma exposure. Whitepaper, squeezemetrics.com, 2017. URL https://squeezemetrics.com/monitor/download/pdf/The_Implied_Order_Book.pdf.
- Aurelio Vasquez, Diego Amaya, Neil D. Pearson, and Pedro Angel Garcia-Ares. 0DTE index options and market volatility: How large is their impact? Technical Report 5113405, SSRN, 2025. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5113405.